Clean Mobility · Data Center Power

Liquid Hydrogen Storage & Fuel Delivery Systems

for Class 8 Trucks and Distributed Power

September 2026





산업통상자원부 지정
수소전문기업
Hydrogen specialized enterprise

LIBECO E&C | PLANHY INC.

1. Company

Overview

Company	LIBECO E&C Co., Ltd. · PLANHY INC. (U.S.)
Founded	February 2021
CEO	Yongseok Kim
Business	Liquid Hydrogen Fuel Storage & Delivery Systems (LHFS) for heavy-duty mobility and distributed power

산업통상자원부 지정 수소전문기업 Hydrogen specialized enterprise

Patents & Certifications

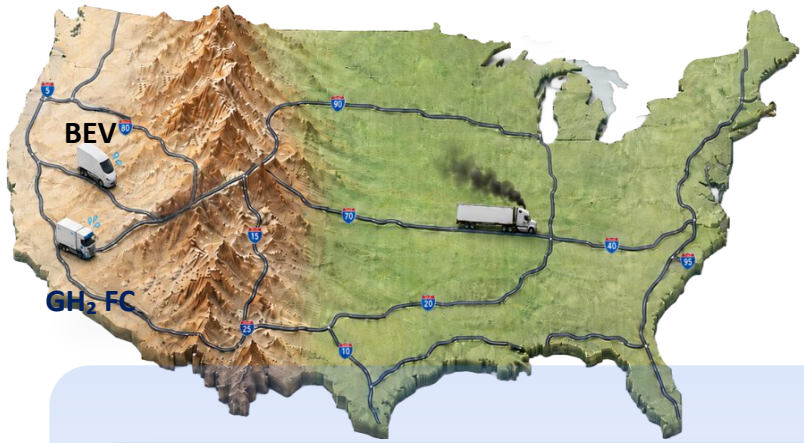
Category	Count	Highlights
▪ Patents ▪ PCT	▪ 7 ▪ 2	▪ Fuel cell system using liquid hydrogen cold energy ▪ Fuel cell system using boil-off gas (BOG) recovery ▪ Liquid hydrogen refueling system ▪ LH_2-based fuel cell powertrain system (granted, PCT) ▪ High-pressure buffer-tank H_2 supply system for FC trucks
▪ Certifications	▪ 5	▪ Venture Company (MSS, 2022) ▪ Hydrogen Specialized Enterprise (MOTIE, 2024) ▪ Certified Startup Company (MSS, 2024) ▪ Corporate R&D Center (MSIT, 2024) ▪ Climate-Tech Company (Gyeonggi Province, 2025)

Milestones

- **2021** • Founded in Korea
- **2022** • Certified Venture Company
- **2024** • Hydrogen Specialized Enterprise (MOTIE)
  - U.S. subsidiary PLANHY INC. established
  - TIPS deep-tech R&D program selected
  - First institutional round closed
- **2025** • Climate-Tech Company program (Gyeonggi)
  - National hydrogen R&D grant (H2KOREA)
  - Scale-up Package + matched investment
  - Korea VC Investment Corp. (KVIC) investment
- **2026** • Global Collaboration — Air Liquide
  - Overseas demonstration program selected
  - Follow-on: NextG Investment, KODIT
  - KIAT Global Talent Program — Phase 1
  - Cumulative funding: KRW 1.63 B (~USD 1.2 M), 5 investors

2. Why Now

North American Class 8 Trucks



Zero-emission truck mandates (ACT)

Cars are going electric – heavy-haul trucks still need an answer

Heavy trucks: ~25% of U.S. transport emissions

Long grades, long routes, autonomous-ready

Solution

LH_2 Fuel Cell Class 8 Truck

Core hydrogen supply system: dual tank, integrated control, U.S. validation

700 bar GH_2
- ✔ Low energy density
- ✔ Packaging issues
- ✔ High-pressure risk
- ✔ Equipment durability

vs.

Liquid H_2
- ✔ High energy density
- ✔ Packages easily
- ✔ Low-pressure storage
- ✔ Simple station ops

Data Center Power



Data centers are power-constrained

Grid capacity shortage → shift to distributed generation

Fast-start backup and bridging power required

Diesel backup draws growing regulatory fire

Solution

LH_2 Fuel Cell Distributed Power

Core hydrogen supply system: multi-tank, integrated control, modular

3. Target Markets

One Platform, Two Core Markets

LIBECO LHFS core technology: LH_2 storage + BOG management + pressure control + heat exchange, integrated

- ✔ **Platform: works with any PEM fuel cell system — Class 8 trucks, distributed power, other heavy mobility**
- ✔ **Modular: scales from 200 kW to 1 MW**

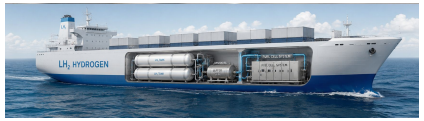


Platform expansion: entering distributed power and heavy hydrogen mobility in parallel

Application	Time to Market	Regulatory Barrier
Distributed Power (data center & building backup)	**2–3 years**	**Low**
Class 8 Trucks	**2–3 years**	**Vibration / crash certification**
Rail (locomotives)	5–7 years	Medium
Marine / Aviation	8–10+ years	Very high





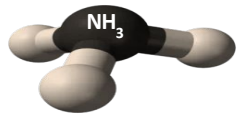


Why distributed power first

- ✔ Demand is visible today — U.S. data center power constraints; proven precedents (Microsoft, ECL, Toyota, Honda)
- ✔ Built on automotive-grade engineering; stationary systems clear a lower certification bar than vehicles

4. Target Segment ① Heavy-Duty Mobility

Clean energy carriers for heavy-duty transport

	Battery	Compressed H_2 (700 bar)	Ammonia (liquid, –33 ℃)	Solution: Liquid Hydrogen (–253 ℃)
				
Storage	Low energy density Cycle-life degradation Fire risk	700 bar compression **Large storage volume**	High energy density	**Utility-scale storage**
Transport	Needs grid buildout Transmission losses	No bulk transport Explosion risk	Bulk transport possible **Toxic leak risk**	**Bulk transport** **Safer than 700 bar**
Fueling & Use	Slow charging, short range Unfit for heavy duty	High compression energy Unfit for heavy duty	Uses existing infrastructure **Costly full-cycle fuel** **(synthesis–liquefy–crack)**	**4× faster fueling than GH_2** **Best fit for heavy duty**
Pros · Cons	Uses existing grid Slow charge, low efficiency Unfit for heavy long-haul	Low well-to-wheel efficiency **Unfit for long-haul**	Easy bulk transport **Costly, low efficiency** **Toxic (NH_3), NO_x emissions**	**High well-to-wheel efficiency** **Heavy payload, long haul** **Needs cryogenic expertise**

□ Primary mobility target: North American Class 8 long-haul, heavy-payload freight

4. Target Segment ② Data Center Backup Power

Data center / backup generation — market size and outlook

31 GW (5,400 sites) 2025 — North America installed capacity	**90 GW+** 2030 outlook (McKinsey)	**106 GW+** 2035 outlook — North America
15,000 units North America diesel gensets (1–4 MW)	**22,000 units** 2030 outlook — diesel + clean	**38,000–48,000** 2035 — North America, clean-power led

PEM hydrogen fuel cells: the leading diesel replacement

- **Diesel genset buildout is exploding**
  : Amazon Manassas — 93 units × 2.5 MW | Oregon permits ~6 GW
- **Local pushback and installation bans**
  : PM / NO_x emissions → community resistance
  : BAAQMD* challenges

- **Microsoft**
  : pledged to eliminate diesel by 2030
  : 3 MW PEM validated → 10+ sites expected by '28
- **Google, Meta, AWS**
  : net-zero commitments create structural diesel-replacement demand

+

- **Aging diesel fleet replacement**
  : 20-year life → clean replacement
  : ~4,000 units by 2030, ~8,000 by 2035

BAAQMD : Bay Area Air Quality Management District

5. Core Technology ① LHFS

Liquid Hydrogen Storage & Fuel Delivery System (LHFS)



LH_2 Storage System

Vehicle-mounted storage system (Gen 1)

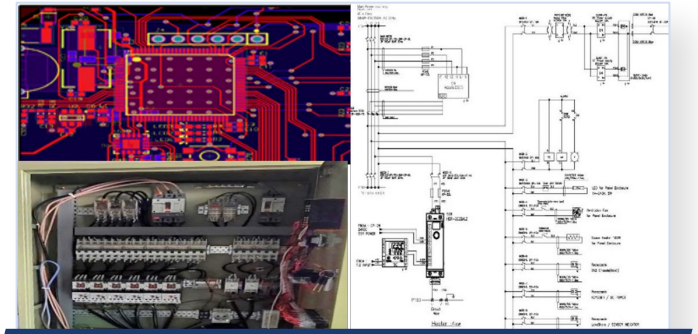
- Lightweight, vehicle-ready design from structural / thermal analysis
- Sub-cooled refueling improves fill performance



LH_2 Delivery System

Energy-maximizing hydrogen delivery

- LH_2 cold energy reused for fuel cell cooling
- BOG energy capture and conversion to power



Integrated Control System

Integrated supply / pressure / thermal control

- Load-following control based on real duty cycles
- Continuous vehicle, efficiency and H_2 system monitoring

Simulated efficiency gains

System energy efficiency up to +7%p

Cold energy +4%p / BOG-to-power +3%p

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Fuel cell power conversion ~+10%

+5%p (45% → 50% conversion)

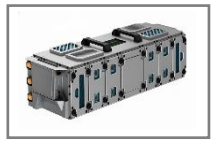
Range ~10 %

5. Core Technology ② LHFS for Class 8 Trucks

North American Class 8 fuel cell powertrain architecture

Integrated LH_2 – fuel cell – battery design with active energy flow control

High Voltage Battery

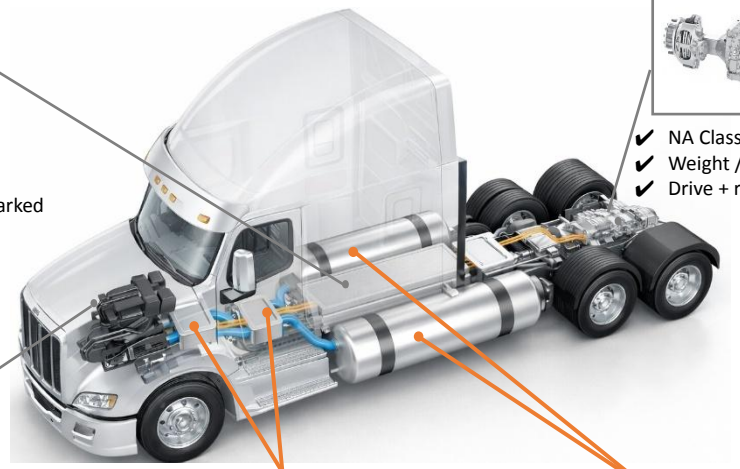


- ✔ Power assist
- ✔ Regen energy storage
- ✔ BOG-to-power when parked

eDrivetrain



- ✔ NA Class 8 rated
- ✔ Weight / power optimized
- ✔ Drive + regen braking

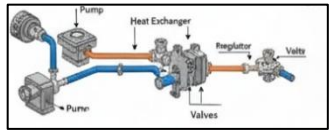


Fuel cell system



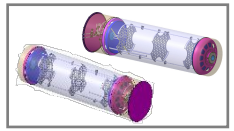
- ✔ NA Class 8 rated
- ✔ Primary vehicle power
- ✔ Cooled by LH_2 cold energy

H2 Supply System



Regulates temperature,pressure, and flow

sLH2 storage system



Tank capacity 100 kg (2 Tanks)

PERFOR-MANCE

✔ **Built for long-haul, heavy-payload duty**
- Meets NA range, sustained-grade and climate requirements
- Sleeper cab retained for Class 8 marketability

EFFICIENCY TCO

✔ **Vehicle integration of the LHFS core technology**
- Vaporization, delivery, cold-energy and BOG modules
- Energy-maximizing control → better payload economics, TCO
- Low-pressure operation cuts station OPEX and fuel cost

SAFETY & REG.

✔ **Zero-venting BOG, safety by design**
- BOG captured and converted to power for up to 20 days
- Meets U.S. GTR No.13 hydrogen emission requirements
- Low-pressure operation improves vehicle / station safety

ICT UPTIME

✔ **Advanced connectivity and digital twin**
- Continuous vehicle and H_2 system monitoring
- AI thermal-flow fault prediction → minimized downtime

Sleeper cab enabled (no GH_2 backpack) | TCO advantage | 20-day BOG capture | Setting the low-pressure H_2 standard

5. Core Technology ③ LHFS for Data Center Power

LIBECO system configuration

PEM output	1 MW class
Backup duration	48 hours
LH_2 required	~3.5 tonnes
Storage	Modular multi-tank
Core functions	LH_2 storage + BOG management + vaporization, heat exchange, flow control

Utility-scale LH_2 Storage-to-PEM Fuel Cell System

PEM fuel cell proof points in data center power

Toyota	**1 MW stationary PEM system co-developed with NREL**
Honda	**2025 — data center power demo using automotive FC**
Microsoft	**2023 — 1.5 MW LH_2-fed PEM demo, 48-hour load test**
ECL	**1 MW PEM as a data center's primary power, commercial** operation since 2025

- ✔ **Automotive fuel cell suppliers (Toyota, Honda)**
  → **stationary demos complete, now entering the market**
- ✔ **Hyperscalers have validated 1 MW-class fuel cell systems**
  → **Microsoft and ECL proved PEM as data center power**

Entry strategy: partner with the fuel cell suppliers already moving into data centers, and own the upstream position — liquid hydrogen storage and delivery.

6. Business Model ① U.S. Market Entry — Ecosystem

H2 Ecosystem Value Chain

RHA* / H_2FCP connect state governments, public and private investors, hydrogen producers, truck OEMs and offtakers (logistics fleets, data centers) into one working supply chain**

Green Hydrogen Production



Air Liquide

Hydrogen Liquefaction



Liquefaction for Storage and Transportation

Air Liquide

LH_2 Transport



High energy density for road transportation

HYROAD

Fueling Station /Dispensing



No Compression
Low Maintenance

FIRST ELEMENT FUEL NIKKISO

LH_2 System & Truck Deploying





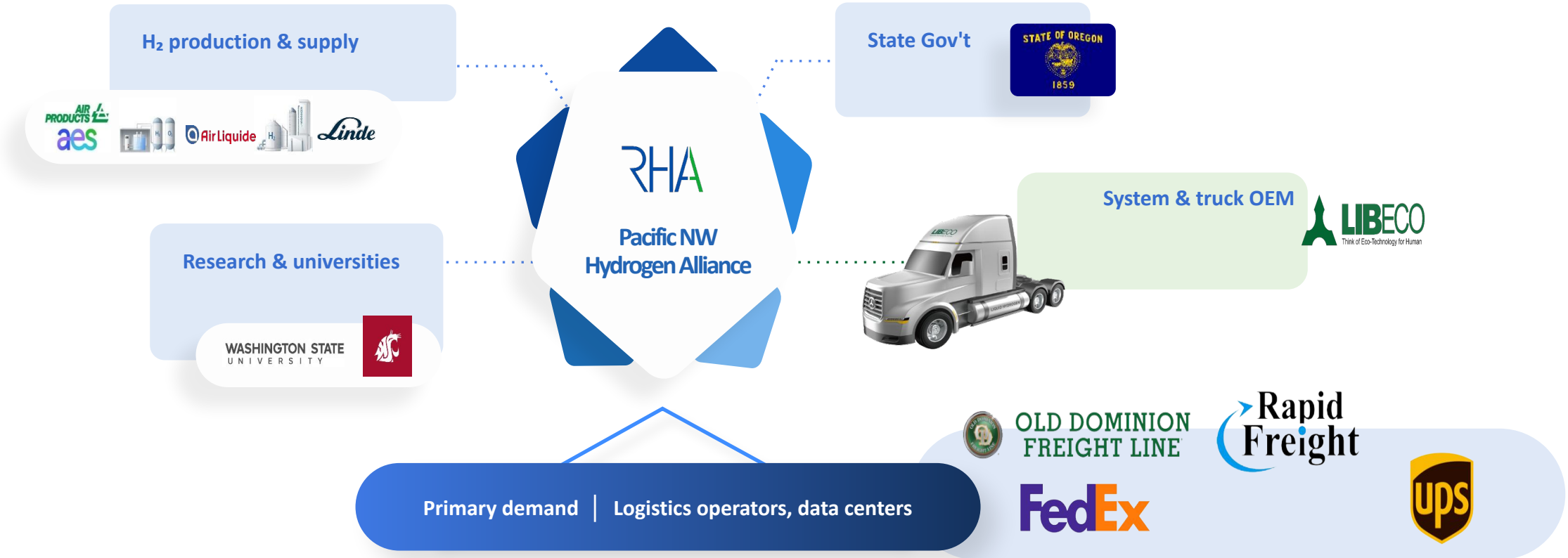
LIBECO Core Business

* RHA : Renewable Hydrogen Alliance (Pacific Northwest) ** H2FCP : Hydrogen Fuel Cell Partnership (California public–private consortium)

6. Business Model ① U.S. Market Entry — Regional Coalition

Regional Clean Transportation Initiative (CTI)

Starting in Oregon, expanding to CA, NJ and TX



※ May 2026 — Full membership in the Renewable Hydrogen Alliance (RHA) confirmed

6. Business Model ② Market Expansion Roadmap

Distributed Power

- Enter the diesel backup replacement market
- Tie into the Pacific Northwest (PNW) LH_2 supply chain
- Partner with fuel cell OEMs such as Toyota and Honda
- Build on the 1 MW reference to supply primary power for smaller data centers

Class 8 Trucks

- Third-party sales of the LHFS platform
- Volume production of LHFS-equipped Class 8 trucks
- Commercial sales to 3PLs, port logistics operators and fleet owners
-

All Hydrogen Mobility

- Scale economics across the LHFS platform
- Mobility market expansion
  - ✔ Logistics & construction equipment
  - ✔ Rail: replacing diesel locomotives
  - ✔ Marine: coastal vessels, port decarbonization mandates
  - ✔ Aviation: UAM, AAM

7. Vision

- ☑ Own the liquid hydrogen fuel system technology for the heavy transport markets where LH_2 is the right answer
  - Higher system maturity and reliability, a widening technology lead, export growth, and global engineering talent
- ☑ Prove the efficiency and economics of LH_2 powertrains in Class 8, then lead the shift to a full hydrogen economy
  - Replace the round-trip inefficiency of 700 bar gaseous hydrogen and extend LH_2 fuel systems across industry at scale



Road Mobility

Rail, specialty vehicles and heavy mobility, then SUV and passenger



Power (Data Centers)

The core LHFS market — fast development to take and hold the U.S. market



Marine

Cryogenic insulation, loading and unloading, BOG handling — core LH_2 technologies in development



Drone



UAM



Defense



Construction

One Big Green PLANET
by Hydrogen



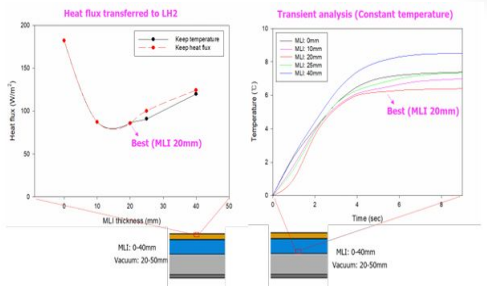
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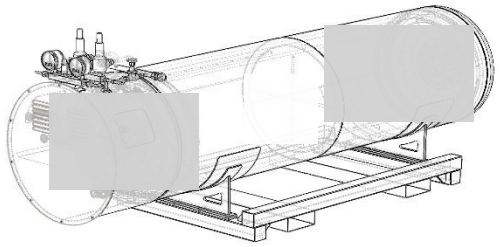
Gen 2 LH_2 Storage Vessel — Build Complete

LH_2 tank design / analysis / fabrication / testing | Major insulation and refueling gains vs. Gen 1 (with BP)

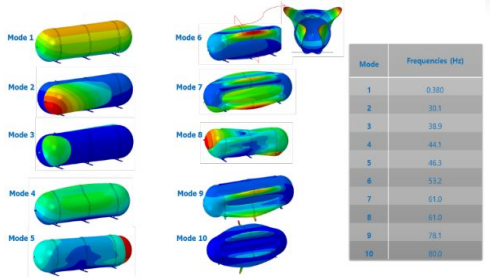
MLI & vacuum design



Vessel design (3D)



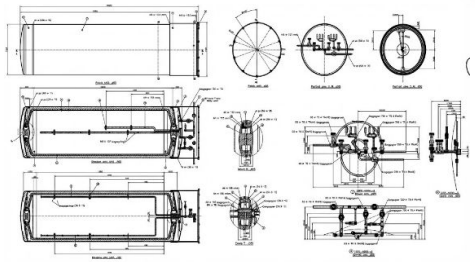
Vibration analysis



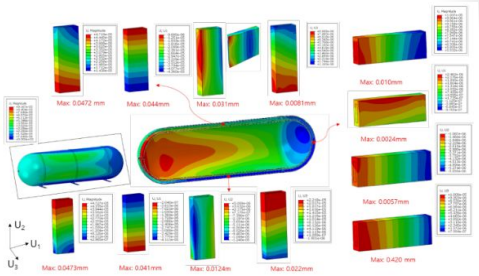
Gen 2 tank fabrication



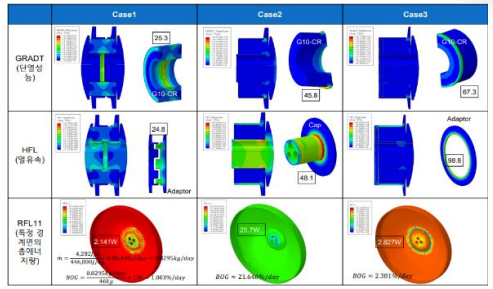
Vessel design (2D)



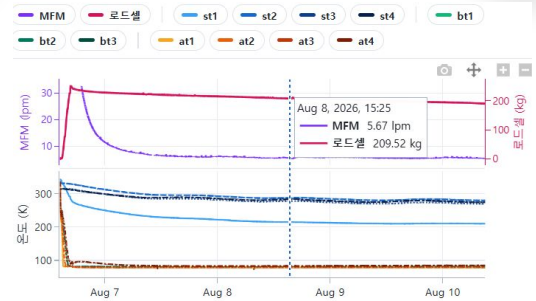
Insulation analysis



Support heat transfer



Gen 2 tank LN_2 test



Gen 2 validation — LN_2 test completed in Korea (Aug 2026) → LH_2 test at a U.S. national laboratory (Oct 2026)